Note on the Dataset — European Jobs

Project: EELISA Data Science Mission

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Date: December 10, 2025

The dataset job_offers.csv was delivered to EELISA on 10.12.2025 for research purposes as part of a broader initiative to analyse labour market trends, digital and green competencies, and emerging skill requirements across Europe. It compiles job postings collected from multiple online recruitment platforms and harmonised to support cross-country comparison.

This document provides an overview of the dataset's structure, the methodology used to collect and normalise the information, and the key variables included. It also outlines the limitations inherent to web-scraped labour data, the assumptions applied during preprocessing, and recommendations for safe and rigorous use of the dataset in research and policy analysis.

The objective of this note is to ensure transparency, reproducibility, and a clear understanding of the dataset's scope so that it can be reliably used by researchers, academic partners, and stakeholders involved in the EELISA Data Science Mission.

Python project files, archives and final datasets can be found on my public GitHub repository : `https://github.com/Antoinechss/EELISA-data-analysis.git`

1 Scraping Methodology

Job postings were retrieved using Python from the EURES platform using its public REST API. A scraper iterated over predefined country codes and keyword queries, sending structured JSON POST requests and parsing paginated responses. Each API response was processed to extract core fields, clean HTML content, and normalise metadata. The scraper handled rate limiting through timed delays, avoided duplicates by tracking job IDs, and incrementally saved results to a CSV file.

To avoid over-representing any single discipline and to request technical jobs, postings were sampled using a balanced keyword strategy. A predefined set of keywords was constructed to cover major domains relevant to the project.

Domain	**Keywords**
Engineering	engineer, engineering, mechanical, electrical, automation, robotics, mechatronics
Digital & Computing	software, developer, programmer, cloud, devops, cybersecurity, network
Data, AI & Machine Learning	data, data science, data analyst, machine learning, ai, artificial intelligence, deep learning
Research & Academia	research, scientist, laboratory, PhD, postdoc, quantitative, mathematics
Sustainability & Green Transition	sustainability, renewable, solar, wind, energy, environmental, climate
Industry & Advanced Manufacturing	manufacturing, industrial, quality, chemistry, materials, biomedical
Business & Innovation	innovation, product manager, project manager

The scraper queried the EURES API separately for each keyword and collected an equal number of postings per term. This ensured a more even distribution of job offers across thematic areas, preventing bias toward any dominant sector in the dataset.

2 Sources

The job postings were collected from **EURES, the European Employment Services portal**. EURES is an official initiative of the European Commission designed to facilitate labour mobility across EU and EEA countries. The platform aggregates job vacancies published by national public employment services and authorised employers, offering a centralised and standardised source of labour market information across Europe. It provides a reliable foundation for cross-country analysis due to its wide geographic coverage, consistent data structure, and direct links with national employment systems. As Turkey was not present on EURES, Turkish jobs were scraped independently on **Yenibiris** job portal. Yenibiris is one of Turkey's leading online recruitment platforms, widely used by companies across sectors to advertise job vacancies and reach qualified candidates. Established in 2000, the platform hosts listings from both large national employers and international firms operating in Turkey. Yenibiris provides structured job advertisements with detailed descriptions, required qualifications, and employer information, making it a valuable source of labour market data. Its broad sectoral coverage and high posting volume allow it to serve as a reliable complement to EURES for capturing job opportunities within the Turkish labour market, which is not represented in the European portal. The keyword scraping methodology described in Section 1 was identical for both data sources.

3 Content

The dataset features a total of 5460 technical/engineering job posts from European countries, with an almost majority coming from EELISA member countries. The per-country distribution is as follows :

Country	Jobs Retrieved
Austria	227
Belgium	244
Bulgaria	94
Croatia	94
Cyprus	72
Czechia	182
Denmark	232
Estonia	76
Finland	166
France	510
Germany	461
Greece	39
Hungary	22
Ireland	83
Italy	288
Latvia	94
Lithuania	91
Luxembourg	205
Malta	104
Netherlands	220
Norway	135
Poland	115
Portugal	72
Romania	68
Slovakia	121
Slovenia	128
Spain	156
Sweden	234
Switzerland	392
Turkey	535

Table 1: Number of job postings retrieved per country

For each job post, the following characteristics are parsed : **job title, date, company name, country, region, full description** . A primary key column job_id is added in the format of the NUTS code of the corresponding country + the number of the job post. For future maps, the coordinates of the job were also added as well as the NUTS code (1 or 2 depending on precision provided on the post). Finally, the activity field is inferred and added in an additional column (see below for detailed inference methodology).

3.1 Locations

The dataset contains job postings from a broad set of European countries, including Belgium, Bulgaria, Czechia, Denmark, Germany, Estonia, Ireland, Greece, Spain, France, Croatia, Italy, Cyprus, Latvia, Lithuania, Luxembourg, Hungary, Malta, the Netherlands, Austria, Poland, Portugal, Romania, Slovenia, Slovakia, Finland, Sweden, Norway, and Switzerland. Notably, around 45% of all postings originate from countries that are part of the EELISA alliance—namely France, Hungary, Romania, Germany, Italy, Turkey, and Switzerland—reflecting the strategic focus of the initiative while maintaining wide European coverage.

Additional location information returned by the API was mapped to NUTS1 and NUTS2 (if provided) regional

Figure 1: Map of countries of interest (EELISA countries in Cyan

codes, allowing each job offer to be associated with a specific region within its respective country. job offers missing information about a specific region within the country were by default assigned to the capital.

3.2 Dates

To ensure that the dataset reflected the most recent labour market trends, I applied a temporal filter based on the distribution of publication dates. The average posting date was 12 October 2025, with timestamps ranging from 3 January 2025 to 20 November 2025. All job offers dated before January 2025 were removed, as they fell outside the relevant time window and contributed little to current labour market analysis. This step ensured that only contemporary and meaningful postings were retained for subsequent processing.

3.3 Activity fields

Activity field — from the 'field' column — was inferred from the job description and the job title using the GPT-4.1 mini model API. The Large Language Model was used as a classification engine to relate each job to one of the following fields :

	Job Field
1	Data Science & Analytics
2	Artificial Intelligence & Machine Learning
3	Software Engineering & Development
4	Cybersecurity & Information Security
5	Cloud, DevOps & Systems Engineering
6	Telecommunications & Networking
7	Internet of Things (IoT) & Embedded Systems
8	Robotics & Automation
9	Electrical & Electronics Engineering
10	Mechanical Engineering
11	Civil & Structural Engineering
12	Chemical & Process Engineering
13	Materials Science & Materials Engineering
14	Aerospace & Aeronautical Engineering
15	Energy Engineering (Nuclear, Oil & Gas)
16	Environmental & Sustainability Engineering
17	Earth & Geosciences
18	Biomedical & Biotech Engineering
19	Bioinformatics & Computational Biology
20	Manufacturing Engineering & Industry 4.0
21	Supply Chain, Operations & Industrial Engineering
22	Product Design, UX, and HCI
23	Quantum Computing & Quantum Engineering
24	Extended Reality (AR/VR/XR) & Immersive Technologies
25	Computer Graphics, Vision & Multimedia
26	Automotive & Transportation Engineering
27	Marine & Naval Engineering
28	Mining & Metallurgical Engineering
29	Agritech & Food Engineering
30	Research & Academia
31	Renewable Energy Engineering

Table 2: List of Technical Job Fields

This field parsing step was also used to remove any non-technical jobs with a field classified as 'Non technical'. Results were then checked manually before removing these potential outliers.

3.4 Duplicates

A number of job postings appeared several times on the platform, sometimes under different identifiers but with identical or very similar content. To handle this, I first applied a coarse deduplication rule by removing all entries sharing the same title, company, and full description. Because some duplicates differed only by minor wording changes, I then performed a second, more fine-grained similarity check. Using the **RapidFuzz library** (token set ratio), I computed a similarity score based on title, company name, and description. A 95% threshold was selected, which effectively identified and removed nearly all remaining duplicates.

3.5 Translations

Translation played a central role in the processing of this dataset. Since the analysis relies heavily on semantic interpretation—particularly the identification of keywords, skill formulations, and competence patterns, it was essential to obtain translations that remained faithful to the original text. The shift from various European languages to English had to be performed with great precision so as not to distort meaning or introduce bias into the final

results.

Content was translated from various European languages to default English using the **GPT-4.1 mini model**. This model has a good balance between translation quality, speed, and cost. Unlike open-source models such as NLLB, which produced inconsistent and sometimes reformulated outputs, gpt-4.1-mini offers literal translations which was a crucial requirement for short content such as job titles. The model should not interpret the meaning of the job title but rather translate it in a literal manner. The model is cost-effective: at approximately 0.15€ per million characters, translating the full dataset (of size 16.95M characters) cost around 2.5€, compared with 300–400€ if we were to use Google Cloud translate API or DeepL. This makes gpt-4.1-mini the most accurate and economically viable solution for large-scale multilingual job market data translation. Details on the translation of the two columns are provided in the next two subsections

3.5.1 Translation of Job Titles

This section is where the least interpretation was to be made by the translation engine. In order to receive following prompt instructions were delivered to the system : *You are a translation engine specialized in job market terminology. Translate all input text into English. Keep the translation short, literal, and strictly aligned with the original meaning. Do not summarize or explain. Do not change the structure. Do not infer missing words. Do NOT output anything except the translated text.*

3.5.2 Translation of Job Full Descriptions

This column in where the majority of characters were contained (16.6 million characters). Time and cost management. The following instructions were delivered to the system : *You are a professional translation engine specialized in multilingual job-market texts. Translate the following job descriptions into English. Keep the translation literal, accurate, and faithful to the original meaning. Preserve structure, bullet points, and formatting as much as possible. Do not summarize. Do not rewrite or improve the text. Do not omit anything. Output only the translations, in the same list order as the inputs.*

4 Documentation

- **EURES Job Portal**: `https://europa.eu/eures/portal/jv-se/home?lang=en`
- **Yenibiris Turkish Job Portal**: `https://www.yenibiris.com/`
- **RapidFuzz Similarity Score Library**: `https://pypi.org/project/RapidFuzz/`
- **OpenAI GPT-4.1 mini model**: `https://platform.openai.com/docs/models/gpt-4.1-mini`
- **Nomenclature of territorial units for statistics (NUTS)**: `https://ec.europa.eu/eurostat/web/nuts`